Align Pitch Observability: Current Model and Proposed Direction

Codex

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1 Problem

On real log `ubx_raw_20260328_153757.bin`, the align filter shows a persistent down/pitch-like error of about 3°. The current filter can initialize and refine yaw reasonably, but pitch has very weak online observability after bootstrap. The result is:

- pitch/down error can persist for long periods,
- gravity updates are too sparse to remove it,
- direct longitudinal scalar pitch corrections are unstable.

2 Notation

- b: body / IMU sensor frame.
- v: vehicle frame.
- q_{vb}: quaternion mapping vehicle frame to body frame.
- $C_{bv}(q_{vb})$: rotation matrix mapping vehicle-frame vectors into body frame.
- f_b: accelerometer specific-force measurement in body frame, i.e. the body-frame accelerometer vector used by align.
- a_{long}: longitudinal acceleration inferred from GNSS velocity change in vehicle coordinates.
- a_{lat}: lateral acceleration inferred from GNSS velocity change in vehicle coordinates.
- g: gravity magnitude.
- $\delta\phi, \delta\theta_p, \delta\psi$: small roll, pitch, and yaw misalignment errors.
- b_a: accelerometer bias term.

3 Current Align State

The align state is a small mount-misalignment error state

$$\delta\theta = \begin{bmatrix} \delta\phi & \delta\theta_p & \delta\psi \end{bmatrix}^T$$

with covariance $P \in \mathbb{R}^{3\times 3}$, where:

- $\delta\phi$: roll misalignment,
- $\delta\theta_p$: pitch misalignment,
- $\delta\psi$: yaw misalignment.

The nominal mount quaternion is q_{vb}, mapping vehicle frame v to body/IMU frame b.

4 Current Observation Model

4.1 Stationary gravity

During stationary windows, the filter low-pass filters gravity in body frame and applies gravity observations on the accelerometer channels:

$$f_b \approx C_{bv}(q_{vb}) \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix}$$

with yaw masked out. This gives roll/pitch observability from gravity.

This is mathematically sound, but only when the vehicle is quasi-static and the gravity direction is representative of the vehicle frame.

4.2 Dynamic horizontal motion

For each GNSS interval, the code computes GNSS horizontal acceleration in vehicle coordinates:

$$a_n = \frac{v_{n,k} - v_{n,k-1}}{\Delta t}, \quad a_{\text{long}} = \hat{t}^T a_n, \quad a_{\text{lat}} = \hat{\ell}^T a_n$$

where $\hat{t}$ is the unit tangent from midpoint GNSS velocity and $\hat{\ell}$ is the lateral unit vector.

It also computes a body-side horizontal specific-force vector by removing gravity projection:

$$f_b^{\text{horiz}} = \bar{f}_b - (\bar{f}_b^T \hat{g}_b)\hat{g}_b$$

then rotates that into vehicle frame:

$$f_v^{\text{horiz}} = C_{vb}(q_{vb})^T f_b^{\text{horiz}}$$

The current dynamic residual is then reduced to a signed horizontal heading error:

$$e_\psi = \text{atan2}\left(f_{v,x}^{\text{horiz}} a_{\text{lat}} - f_{v,y}^{\text{horiz}} a_{\text{long}},\ f_{v,x}^{\text{horiz}} a_{\text{long}} + f_{v,y}^{\text{horiz}} a_{\text{lat}}\right)$$

and applied as a yaw-only correction:

$$\delta\psi \leftarrow \delta\psi - K_\psi e_\psi$$

4.3 Turn gyro

During turn-valid windows, gyro residuals provide some additional tilt support. In practice this contributes a small amount to pitch, but not enough to remove the persistent 3° offset.

5 Why Pitch is Weakly Observable in the Current Design

The current dynamic branch is designed primarily for yaw:

- gravity updates affect roll/pitch only,
- horizontal dynamic updates affect yaw only,
- turn-gyro gives a small secondary tilt correction.

So after bootstrap, pitch has no real drive-time correction path except:

- sparse stationary gravity updates,
- small turn-gyro effects.

That matches the observed diagnostics on `153757`: horizontal updates contribute essentially zero pitch correction, while gravity contributes very little total pitch correction over the whole drive.

6 Why the Simple Scalar Pitch Fix Failed

The attempted scalar idea was effectively:

$$r_{\text{long}} = a_{\text{long,gnss}} - a_{\text{long,imu}}$$

with a small-angle interpretation

$$r_{\text{long}} \approx g\,\delta\theta_p + b_{a,x}$$

If that model were valid and the sign were consistent, then a slow per-window correction and a batch estimate would converge to the same answer.

But on the real log, the residual is not consistent:

- straight accel windows have one mean sign,
- straight brake windows have the opposite mean sign,
- a single constant longitudinal bias does not fit the data.

So the residual is not a stable signed measurement of pitch. It is maneuver-dependent. Because of that:

- per-window updates chase inconsistent residuals,
- batching only averages inconsistent residuals,
- lowering gain only slows the wrong result.

7 Proper Mathematical Direction

The correct way to introduce dynamic pitch observability is to use the full specific-force vector model, not a scalar heuristic.

7.1 Vector specific-force model

For straight longitudinal motion, the ideal vehicle-frame specific force is approximately

$$f_v^\star \approx \begin{bmatrix} a_{\text{long}} \\ 0 \\ -g \end{bmatrix}$$

and the measured body-frame force is

$$f_b \approx C_{bv}(q_{vb}) f_v^\star + b_a + \eta$$

where b_a is accelerometer bias and η is noise/model error.

This is the right measurement equation because pitch changes the *vector geometry*, not just the scalar longitudinal component.

7.2 Linearized residual

Define

$$r_f = f_b - C_{bv}(q_{vb}) f_v^\star$$

Then linearize with respect to small misalignment:

$$r_f \approx H_f \, \delta\theta + b_a + \eta$$

where H_f comes from the skew-symmetric Jacobian of the rotated vector.

For straight-line windows, the most useful component is the part of H_f that makes pitch observable. This should be used with:

- strong gating on straightness and low lateral contamination,
- explicit handling of maneuver phase,
- either an estimated bias term or a nuisance-robust weighting scheme.

8 Recommended Estimator Structure

The next version should not do "one window → one pitch correction." Instead:

8.1 1. Build a buffered straight-motion pitch estimator

Accumulate a buffer of straight accel/brake windows and solve a small weighted least-squares problem over the vector residual:

$$\min_{\delta\theta_p} \sum_{i \in \mathcal{S}} w_i \left\| f_{b,i} - C_{bv}(q_{vb} \oplus \delta\theta_p) \begin{bmatrix} a_{\text{long},i} \\ 0 \\ -g \end{bmatrix} \right\|^2$$

or its linearized equivalent.

8.2 2. Use conservative gating

Only include windows with:

- low course rate,
- low lateral acceleration,
- sufficient speed,
- sufficient longitudinal acceleration magnitude,
- stable sign/consistency over the buffer.

8.3 3. Feed only a small bounded correction into align

Once the buffered estimate is consistent, inject only a small pitch correction step into the align state. This preserves smoothness and avoids sharp maneuver-to-maneuver flips.

8.4 4. Keep yaw and pitch logically separated

Yaw should still be primarily turn-observed. Pitch should be straight-motion observed. The current formulation mixes dynamic cues in a way that is good for yaw but too weak for pitch.

9 Summary

The current align formulation is mathematically suited for:

- gravity-based roll/pitch initialization and maintenance,
- dynamic yaw correction from horizontal motion,
- small additional tilt support from turn gyro.

It is *not* designed to provide strong dynamic pitch correction during normal driving.

The proper fix is not to increase gain on the current scalar longitudinal residual. The proper fix is to add a new vector-based straight-motion pitch observation model, buffered and heavily gated, so that pitch becomes observable without destabilizing the filter.